HDOC Day-48 Activity

Pivot Index in a Variable-Length Array Using Pointers

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Aim

To read a variable-length integer array, pass its address to a function named **pivot_index()** using a pointer, and find the leftmost index where the sum of all elements on the left is equal to the sum of all elements on the right.

Requirements

1. Read the size and elements of VLA **arr[]**.
2. Pass the array to **pivot_index()** using its address.
3. Calculate left sum and right sum for each index.
4. For index 0, left sum is 0.
5. For the last index, right sum is 0.
6. Return/display the **leftmost** index where left sum = right sum.
7. Display **-1** when no pivot index exists.

Algorithm

1. Read size and array elements.
2. Calculate the total sum of all array elements.
3. Set leftSum = 0.
4. For every index i from left to right, calculate **rightSum = totalSum - leftSum - arr[i]**.
5. If leftSum == rightSum, immediately return i. This gives the leftmost pivot index.
6. Otherwise add arr[i] to leftSum and continue.
7. If the loop finishes without a match, return -1.

C Program

```
#include <stdio.h>

int pivot_index(int *arr, int size)
{
    int i;
    int totalSum = 0;
    int leftSum = 0;
    int rightSum;

    for (i = 0; i < size; i++)
        totalSum += *(arr + i);

    for (i = 0; i < size; i++)
    {
        rightSum = totalSum - leftSum - *(arr + i);

        if (leftSum == rightSum)
            return i;

        leftSum += *(arr + i);
```

```
    }

    return -1;
}

int main()
{
    int size, i, index;

    printf("Enter size of array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid array size!\n");
        return 0;
    }

    int arr[size];

    printf("Enter %d elements:\n", size);

    for (i = 0; i < size; i++)
        scanf("%d", &arr[i]);

    index = pivot_index(arr, size);

    if (index == -1)
        printf("-1\n");
    else
        printf("Pivot index: %d\n", index);

    return 0;
}
```

Sample Outputs

```
Example 1
Enter size of array: 6
Enter 6 elements:
1 7 3 6 5 6
Pivot index: 3

Explanation:
Left sum  = 1 + 7 + 3 = 11
Right sum = 5 + 6 = 11

Example 2
Enter size of array: 3
Enter 3 elements:
1 2 3
-1

Example 3
Enter size of array: 3
Enter 3 elements:
2 1 -1
Pivot index: 0

Explanation:
Left sum  = 0
Right sum = 1 + (-1) = 0
```

Function Explanation

pivot_index(int *arr, int size): The function first calculates the total array sum. During the second loop, it maintains the sum of elements strictly to the left of the current index. The right sum is found using **totalSum - leftSum - current element**. The first equal pair of sums causes an immediate return, ensuring the leftmost pivot index.

Pointer Concept Used

The array is passed by its base address. Therefore ***(arr + i)** accesses the same element as **arr[i]**. No copy of the whole array is required when calling pivot_index().

Easy Memory Trick

Pivot means BALANCE.
Left side sum = Right side sum.
Right sum formula: Total - Left - Current.
After checking an index: **Left = Left + Current**.

Dry Run for Example 1

Array = 1, 7, 3, 6, 5, 6; Total = 28.
At index 0: left = 0, right = 27 -> not equal.
At index 1: left = 1, right = 20 -> not equal.
At index 2: left = 8, right = 17 -> not equal.
At index 3: left = 11, right = 11 -> **Pivot index = 3**.

Complexity

The array is scanned twice, so the time complexity is **O(n)**. Only a few integer variables are used, so extra space is **O(1)**.

Result

Thus, the variable-length array was passed using a pointer and the leftmost pivot index was found successfully by comparing the left and right sums. The program correctly displays -1 when no pivot index exists.

Quick Viva Questions

1. What is a pivot index? An index where the sum of elements strictly on its left equals the sum strictly on its right.
2. What is the left sum at index 0? 0.
3. What is the right sum at the last index? 0.
4. What formula is used for right sum? totalSum - leftSum - arr[i].
5. Why return immediately after finding equality? To return the leftmost pivot index.
6. What is the pivot index of 1 7 3 6 5 6? 3.